

## Linear Regression using California Housing Dataset

### Objective

To build a Linear Regression model for predicting California housing prices using the California Housing dataset.

### Dataset

- Source: Scikit-learn California Housing Dataset
- Total Records: **20,640**
- Features: **8**
- Target Variable: **Price**

### Exploratory Data Analysis (EDA)

- Checked dataset shape and information.
- Verified missing values (none found).
- Generated statistical summary.
- Plotted histograms for feature distributions.
- Generated a correlation heatmap.

### Model

- Algorithm: **Linear Regression**
- Data Split: **80% Training, 20% Testing**

### Evaluation Metrics

MAE = 0.5332001304956555

RMSE = 0.7455813830127763

$R^2$  Score = 0.575787706032451

### Visualizations

- Histogram
- Correlation Heatmap
- Actual vs Predicted Scatter Plot
- Residual Plot

## **Conclusion**

The Linear Regression model was successfully trained using the California Housing dataset. Exploratory Data Analysis helped understand the data distribution and feature relationships. The model was evaluated using MAE, RMSE, and  $R^2$  Score, demonstrating its effectiveness as a baseline model for house price prediction.